Drake Gonzales

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https://swayngee.github.io/E155-Digital-design-Projects-/ • https://github.com/Swayngee

EDUCATION

Harvey Mudd College - Claremont, CA **Expected May 2027**

Bachelor of Science in General Engineering (Emphasis: Electrical, RF & Computer Engineering)

- Coursework: Radio Frequency Circuit Design, System-on-Chip (SoC) Design, Signal Engineering, Microprocessor Systems, Electric & Magnetic Circuits/Devices, Digital Electronics & Computer Engineering, Applied Math for Engineering, Engineering Systems

EXPERIENCE

Quantum Instrumentation Control Kit (QICK) | HRL Laboratories / IBM **Aug 2026 - Present**

Clinic Program, Harvey Mudd College

- Replacing the QICK tProcessor with a standardized RISC-V-based quantum-control architecture for real-time control of superconducting and spin qubit experiments.
- Collaborating closely with HRL and Fermilab engineers and developers to ensure feature completeness and compatibility with existing QICK and spinQICK APIs, enabling precise timing, waveform generation, and pulse-level control.
- Developing a maintainable, extensible, open-source quantum-control architecture on the RISC-V ecosystem and a Xilinx UltraScale+ RF System-on-Chip (RFSoC) FPGA, supporting high-speed RF/microwave signal generation and real-time qubit control.

GPS IV | The Aerospace Corporation / The FAA **Jan 2026 - May 2026**

Clinic Program, Harvey Mudd College

- Used SOAP (Satellite Orbit Analysis Program) to perform orbital mechanics calculations and model future Laser Inter-Satellite Link (LISL) architectures for GPS IV satellite applications (ITAR).
- Modeled LISL architectures by analyzing orbital geometry, link availability, pointing requirements, and satellite relative motion to identify viable configurations.
- Presented technical findings on orbital geometry, LISL feasibility, and system-level trade-offs to aerospace engineers at The Aerospace Corporation's Concurrent Design Center (CDC).

Student Intern | Roncelli Technologies **Summer 2026**

- Assisted with AI-driven automation, PLC/CNC programming, robotic automation, and process optimization.
- Collaborated with the SMDT team to analyze ERP, IIoT (Industrial Internet of Things), and production flows.

Autonomous Drone Research | Harvey Mudd College **Jan 2025 - Jan 2026**

- Designed a multi-UAV wildfire inspection system using embedded platforms.
- Developed thermal and RGB computer vision data pipelines using a YOLOv8 model, achieving fire detection accuracy of up to 70%.

PROJECTS

RF Circuit Design | Harvey Mudd College **Aug 2026 - Present**

- Designed, simulated (LTspice), built, and VNA-characterized a 5th-order Butterworth low-pass filter (100 MHz), measuring 0.1 dB insertion loss and 0.95 dB ripple; modeled via and pad parasitics (1.4 nH, 0.2 pF) to match simulation to hardware.
- Designed and tuned a PCB L-matching network (50 Ω to 270 Ω at 100 MHz) using SOL-calibrated VNA measurements and Smith charts; characterized transmission-line reflections and VSWR with time-domain measurements.

FPGA & System-on-Chip (SoC) Design | Harvey Mudd College **Aug 2025 - Jan 2026**

- Designed a real-time 3-band audio equalizer on an FPGA that independently boosts or attenuates low, mid, and high frequencies using digital IIR filters and configurable gain controls.
- Designed a pipelined RISC-V processor from ISA-level specifications with forwarding, load-use hazard handling, and a static BTFNT branch predictor, reaching a 4.17 ns clock period and improving CoreMark performance 4.5× over baseline (2,298 vs. 507).

LEADERSHIP & ACTIVITIES

Student Tutor | Microprocessor Systems (E155) | Harvey Mudd College **Aug 2026 - Present**

- Host lab hours for students learning to program FPGAs and microcontrollers.

Mudd Amateur Rocketry Club (MARC) | Harvey Mudd College **Aug 2024 - Jan 2026**

- Co-led the avionics team; recorded the highest-altitude launch in Harvey Mudd history (17,172 ft).
- Led High-Power Rocketry and taught students how to obtain Level 1 certification.

Athletic Director & Gym Head Supervisor | Harvey Mudd College **Aug 2024 - Present**

- Work with the student board, faculty, the President's Cabinet, and the Board of Trustees to allocate student funds.

SKILLS

Programming: SystemVerilog, Verilog, C, RISC-V Assembly, Python, MATLAB, PLC, CNC

RF & Lab Equipment: Vector Network Analyzer, Spectrum Analyzer, Oscilloscope, Logic Analyzer, Amplifiers, Mixers, Oscillators, Frequency Multipliers, High-Power Limiters, Switches, Antennas

Hardware & Platforms: FPGA (Intel Quartus, Questa), Xilinx RFSoC, STM32, LTspice, Jetson Orin, Raspberry Pi, Pixhawk, Soldering, SMT, SolidWorks, Git

Protocols: SPI, I^2C, UART, CAN, Ethernet